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Energy-equidistributed moving mesh strategies for simulating Hamiltonian partial differential equations[☆]

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ABSTRACT

This paper presents an innovative energy-equidistributed moving mesh strategy for simulating Hamiltonian partial differential equations (PDEs) characterized by solitons and rapid temporal variations. A novel framework, named the Energy Equidistribution Principles (EEPs), is introduced, highlighting the critical role of energy conservation in achieving accurate simulations. Building on EEPs, three kinds of energy-equidistributed moving mesh PDEs (EMMPDEs) are proposed, each grounded in different methodologies. These strategies are rigorously examined in terms of their convergence conditions and rates. Both theoretical analysis and numerical experiments demonstrate that the proposed EMMPDEs offer superior robustness and effectiveness in long-term simulations, compared to traditional arc-length-equidistributed MMPDEs.

1. Introduction

Hamiltonian PDEs are widely applied in fields such as meteorology, quantum mechanics, and nonlinear optics. Examples include the Sine–Gordon, Schrödinger, KdV, and Maxwell equations. In this paper, the theoretical discussion focuses on Hamiltonian PDEs of the form [1]:

$$\frac{\partial \mathbf{u}}{\partial t} = S \frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}], \quad \mathbf{x} \in \Omega \subseteq \mathbb{R}^d, \quad \mathbf{u} = \mathbf{u}(\mathbf{x}, t) \in \mathbb{R}^m, \quad (1.1)$$

where S is a skew-symmetric operator with respect to the L^2 inner product and $\frac{\delta H}{\delta \mathbf{u}}$ represents the variational derivative of a functional $H[\mathbf{u}]$ [2]:

$$H[\mathbf{u}] = \int_{\Omega} \omega(\mathbf{x}; \mathbf{u}^{(l)}) d\mathbf{x}, \quad (1.2)$$

ω is the energy density function, $\mathbf{u}^{(l)}$ denotes function $\mathbf{u}$ and its partial differential derivatives up to l .

If $\mathbf{u}$ satisfies suitable boundary conditions such as null or periodic boundary conditions, the energy H remains invariant over time [3,4]:

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$$\frac{dH}{dt} = \left(\frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}], \frac{\partial \mathbf{u}}{\partial t} \right)_{L^2} = \left(\frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}], S \frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}] \right)_{L^2} = 0.$$

This represents the so-called conservation law, which serves as a key criterion for the success of long-term dynamical simulations.

Traditional fixed-grid methods often struggle with systems exhibiting rapid temporal changes or solitons [5–8]. To address this, moving mesh strategies based on equidistribution principles (EP) by de Boor [9], have been applied to improve mesh adaptability [10,11]. However, existing methods commonly rely on geometric equidistribution (for example, arc-length equidistribution), which may not fully capture the energy dynamics of Hamiltonian systems.

In this paper, we introduce a novel EEP that focuses on energy equidistribution. The EEP clusters mesh points in regions with large energy variations, enhancing accuracy and stability in simulations. Based on this principle, we propose three kinds of energy-equidistributed moving mesh PDEs (EMMPDEs) and demonstrate their effectiveness in preserving energy and improving computational efficiency through numerical experiments.

The structure of the paper is as follows: Section 2 introduces the EEPs. Section 3 presents three EMMPDE strategies. Section 4 discusses the numerical algorithm and theoretical analysis, and Section 5 details the numerical experiments.

2. Energy equidistribution principles (EEPs)

We interpret the idea in one dimensional (1-D) case. Given a function $\mathbf{u} = \mathbf{u}(x, t)$, without loss of generality, one can assume that $x \in [0, 1]$. A one-to-one coordinate transformation between the computational domain and physical domain is defined as:

$$x = x(\xi, t), \quad \xi \in [0, 1], \quad (2.1)$$

subject to the boundary conditions:

$$x(0, t) = 0, \quad x(1, t) = 1. \quad (2.2)$$

Using the chain rule, the original PDE (1.1) transforms into:

$$\frac{\partial \mathbf{u}}{\partial t} - \frac{\partial \mathbf{u}}{\partial x} \frac{\partial x}{\partial t} = S \frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}]. \quad (2.3)$$

Hamiltonian systems are inherently infinite-dimensional, and their numerical solution typically involves spatial discretization, reducing the problem to a finite dimensional Hamiltonian ordinal differential equation (ODE) system. Existing moving mesh strategies for Hamiltonian PDEs predominantly employ geometric equidistribution techniques [2,12,13] based on the EP proposed by de Boor [9], which defines the transformation $x(\xi, t)$ by requiring the equidistribution of a positive monitor function (such as arc-length $\omega = \sqrt{1 + |\mathbf{u}_x|^2}$).

The EP is effective in concentrating knots in areas of significant variation, thereby achieving high accuracy with fewer knots. Based on EP, we propose an energy-based equidistribution principle (EEP) which incorporate the energy conservation property of Hamiltonian systems into the knots movement:

$$E(x(\xi, t), t) \doteq \int_0^{x(\xi, t)} \omega(x; \mathbf{u}^{(l)}) dx - \xi \int_0^1 \omega(x; \mathbf{u}^{(l)}) dx = 0. \quad (2.4)$$

With $\xi_j = \frac{j}{N}$, the energy associated with each knot is equidistributed, i.e., the energy at the j -th knot satisfies:

$$H_j(t) = \frac{1}{2} \int_{x_{j-1}(t)}^{x_{j+1}(t)} \omega(x; \mathbf{u}^{(l)}) dx = \frac{1}{N} H(t), \quad (2.5)$$

where $x_j = x(\xi_j)$. This contrasts with geometric approaches that focus on shape, as the proposed method clusters knots in regions of significant energy variation, improving the accuracy of energy dynamics. For Hamiltonian PDEs, the total energy $H(t)$ is expected to remain conserved over time, meaning $H(t) = H_0$, where H_0 is the initial energy. Incorporating this energy conservation into the monitor function, we redefine the EEP in this paper as:

$$E(x(\xi, t), t) \doteq \int_0^{x(\xi, t)} \omega(x; \mathbf{u}^{(l)}) dx - \xi H_0 = 0. \quad (2.6)$$

This EEP ensures that each ODE system remains Hamiltonian, with constant energy $H_j(t) = \frac{1}{N} H_0$, aligning with the fundamental characteristics of Hamiltonian systems. To avoid the complexities of direct integration, we consider the first and second derivatives of E :

$$E_\xi(x, t) = \omega(x; \mathbf{u}^{(l)}) \frac{\partial x}{\partial \xi} - H_0 = 0, \quad (2.7)$$

$$E_{\xi\xi}(x, t) = \frac{\partial}{\partial \xi} \left(\omega(x; \mathbf{u}^{(l)}) \frac{\partial x}{\partial \xi} \right) = 0. \quad (2.8)$$

EEPs (2.6), (2.7), and (2.8) along with the boundary conditions (2.2) are equivalent, which provide a robust framework for implementing the EEPs. The EEPs ensures the equidistribution of energy across the knots, promoting both stability and accuracy in the solution. This is particularly beneficial for simulating PDEs with solitons and other long-term dynamics.

Remark 2.1. While the monitor function ω in EEPs offers many advantages in moving mesh methods for Hamiltonian PDEs, it is important to note that the positiveness of the monitor function ω is required. To guarantee this, one can change ω into $\omega + C$, where C is a sufficiently large positive constant to ensure $\omega + C > 0$.

3. Energy-equidistributed moving mesh strategies

Due to the nonlinearity of these EEPs, directly calculating the zero points poses significant challenges. The widely used methods include the root-searching iteration method [14,15], the Lagrangian method [16], and the Taylor expansion method [17]. Based on these three approaches, we propose three kinds of EMMPDEs.

3.1. EMMPDEs based on iteration methods

To begin, we recall standard results from root-searching iteration methods. Here we take EEP (2.6) as the example, the method is available for EEP (2.7) and EEP (2.8) also. When t is fixed, the fixed-point iteration method and Newton's iteration method can be expressed as:

$$x_{n+1} = x_n - \lambda E(x_n, t), \quad x_{n+1} = x_n - \frac{\lambda E(x_n, t)}{E_x(x_n, t)}$$

separately, where λ is a constant that controls the iteration speed. For the first method, the convergence condition for this method is: $0 < \lambda E_x(x_n, t) < 2$ and the convergence speed is one (linear convergence). For Newton's method, the convergence condition is $E_x(x_n, t) \neq 0$, and the convergence speed is two (quadratic convergence). By substituting finite differences with derivatives in the t direction, we obtain the first system of EMMPDEs.

The first kind of EMMPDEs for Hamiltonian PDE (1.1) is:

$$\begin{aligned} \text{(EMMPDE 1)} \quad & \frac{\partial x}{\partial t} = -\frac{1}{\tau} \left(\int_0^x \omega dx - \xi H_0 \right), \\ \text{(EMMPDE 2)} \quad & \frac{\partial x}{\partial t} = -\frac{1}{\tau} \frac{\left(\int_0^x \omega dx - \xi H_0 \right)}{\omega}, \end{aligned}$$

where τ is the time step.

Based on the iteration theory, the convergence conditions and speed can be determined. However, these EMMPDEs only approximate the roots of the EEPs when time t is fixed. When the solutions evolve over time, a time lag arises between the equidistributed knots at the new time step. The faster the solution changes, the larger the time lag becomes, which introduces significant limitations to the iterative method - a phenomenon confirmed by our previous experiments [18].

3.2. EMMPDEs based on lagrangian methods

The second approach involves maintaining the original energy distribution during time iterations. This implies that the energy proportion should remain consistent throughout the temporal evolution. To achieve this, one can differentiate the EEPs, for example, EEP (2.6), with respect to time:

$$\omega(x; \mathbf{u}^{(l)}) \frac{\partial x(\xi, t)}{\partial t} + \int_0^{x(\xi, t)} \frac{\partial \omega(x; \mathbf{u}^{(l)})}{\partial t} dx = 0.$$

As a result, we obtain the second kind of EMMPDEs:

$$\text{(EMMPDE 3)} \quad \frac{\partial x}{\partial t} = -\frac{\int_0^x \omega_t dx}{\omega},$$

where ω_t is transformed into spatial differentiation by applying the original PDE (1.1).

These EMMPDEs ensure that the knots preserve the original energy distribution over time. However, they do not adjust the energy proportions or relocate knots to achieve equidistribution. If the initial knots are suboptimal, this imbalance will persist. Additionally, numerical errors in practical calculations can lead to deviations, even if the knots start in ideal positions.

3.3. EMMPDEs based on Taylor expansion methods

Motivated by the above two methods, we aim to construct EMMPDEs that not only drive the system towards equidistribution at each fixed time step but also maintain the energy proportion during the time iterations. In other words, the resulting function $x(\xi, t)$ are supposed to satisfy the EEPs after a short time interval τ , i.e.,

$$E(x(\xi, t + \tau), t + \tau) = 0. \quad (3.1)$$

Suppose E is smooth enough, we expand the equation in a Taylor series at t :

$$E(x(\xi, t), t) + \tau E_x(x(\xi, t), t) \frac{\partial x(\xi, t)}{\partial t} + \tau E_t(x(\xi, t), t) \leq K_2 \tau^2, \quad (3.2)$$

where

$$K_2 = \sup_{x,t} \left\{ \left| \frac{\partial^2 E}{\partial x^2} \right| \left(\frac{\partial x}{\partial t} \right)^2, \left| \frac{\partial^2 E}{\partial x \partial t} \right| \frac{\partial x}{\partial t}, \left| \frac{\partial^2 E}{\partial t^2} \right| \right\}.$$

Neglecting higher-order terms, we obtain the third kind of EMMPDEs for Hamiltonian PDE (1.1):

$$\text{(EMMPDE 4)} \quad \frac{\partial x}{\partial t} = -\frac{1}{\tau} \frac{\int_0^x \omega dx - \xi H_0}{\omega} - \frac{\int_0^x \omega_t dx}{\omega}.$$

EMMPDE 4 is a combination of EMMPDE 2 and 3. It consists of two terms: one drives the knots towards equidistribution, while the other ensures consistency of the distribution over time. Following a similar approach to [17], the asymptotic convergence to equidistribution can be rigorously proven.

4. Moving mesh methods based on EMMPDEs

4.1. Algorithm

For spatial discretization of the PDE (2.3), we employ the centered difference method for all spatial derivatives on the energy contours based on EMMPDEs, the resulting ODE system at $\{x_j\}_{j=1}^{N-1}$ are:

$$\begin{cases} \frac{d\mathbf{u}_j}{dt} - \frac{(\mathbf{u}_{j+1} - \mathbf{u}_{j-1})}{(x_{j+1} - x_{j-1})} \frac{dx_j}{dt} = \left(S \frac{\delta H}{\delta \mathbf{u}}[\mathbf{u}] \right)_j, \\ \frac{dx_j}{dt} = -\frac{1}{\tau} \frac{E_j}{\omega_j} - \frac{(E_t)_j}{\omega_j}, \end{cases} \quad (4.1)$$

where $\xi_j = \frac{j}{N}$, $x_j(t) \approx x(\xi_j, t)$, $\mathbf{u}_j(t) \approx \mathbf{u}(x(\xi_j, t), t)$, etc.

Various temporal discretization methods can be employed to solve the ODE system (4.1) along the energy-equidistributed lines. To better evaluate the performance of EMMPDEs, we utilize MATLAB's *ode15i* solver, which is well-suited for stiff systems, and adaptively adjusts time step sizes while maintaining fixed absolute and relative error tolerances. By enabling larger time steps, *ode15i* enhances computational stability and efficiency, allowing us to comprehensively assess the EMMPDEs' effectiveness, stability, robustness, and long-term approximation capabilities.

4.2. Theoretical analysis

For the continuous energy function H (1.2) and H_j (2.5), we can derive the approximation properties for the discrete energy associated with the moving lines $x_j(t)$. According to the error estimate of the midpoint quadrature rule, there exist two constants C_1 and C_2 , independent of h , such that

$$(H_d)_j := \omega_j \frac{x_{j+1} - x_{j-1}}{2} = H_j + C_1 h^3, \quad (4.2)$$

$$H_d := \sum_{j=1}^N \omega_j \frac{x_{j+1} - x_{j-1}}{2} = H + C_2 h^2, \quad (4.3)$$

where $h = \max_j(x_j - x_{j-1})$, $x_{N+1} = 1 + x_1$ and the constants $C_i = C_i(\|w_{xx}\|_\infty)$, $i = 1, 2$.

Next, we prove that $(H_d)_j$ changes slightly over time. Let $\bar{x}_j$ denote the exact solution of EEPs on $\xi = 1/N$, and assume that x_j remains close to $\bar{x}_j$, i.e. $|x_j - \bar{x}_j| \leq C_0 h^2$, where C_0 is a constant determined by the moving mesh algorithm. Under this assumption, we have

$$\begin{aligned} (H_d)_j &= \frac{1}{2} \int_{x_{j-1}}^{x_{j+1}} \omega(x; \mathbf{u}^{(l)}) dx + C_1 h^3, \\ &= \frac{1}{2} \int_{\bar{x}_{j-1}}^{\bar{x}_{j+1}} \omega(x; \mathbf{u}^{(l)}) dx + \frac{1}{2} \int_{x_{j-1}}^{\bar{x}_{j-1}} \omega(x; \mathbf{u}^{(l)}) dx + \frac{1}{2} \int_{\bar{x}_{j+1}}^{x_{j+1}} \omega(x; \mathbf{u}^{(l)}) dx + C_1 h^3 \\ &= \frac{1}{N} H + C_1 h^3 + C_3 h^2 = \frac{1}{N} H + C_4 h^2, \end{aligned}$$

where $C_3 = C_3(\|w_x\|_\infty)$ and $C_4 \leq C_1 h + C_3$. This leads to the following estimate

$$\frac{d(H_d)_j}{dt} = C_5 h^2,$$

where $C_5 = C_5(hC_3\|\omega_{xxt}\|_\infty, \|\omega_{xt}\|_\infty)$. Furthermore, we obtain $\frac{dH_d}{dt} = \frac{dH}{dt} + C_6 h^2 = C_6 h^2$ due to the energy conservation property ($\frac{dH}{dt} = 0$). Here $C_6 = C_6(\|\omega_{xxt}\|_\infty)$.

5. Numerical experiments

5.1. 1-D Hamiltonian PDE

We are concerned with the numerical solution of 1-D Sine-Gordon equation with the initial value problem:

$$\begin{cases} u_{tt} - u_{xx} + \sin(u) = 0, & x \in \mathbb{R} \times (0, T], \\ u(x, 0) = u_0(x), & u_t(x, 0) = u_1(x), \end{cases} \quad (5.1)$$

with the periodic boundary conditions. The exact solution is:

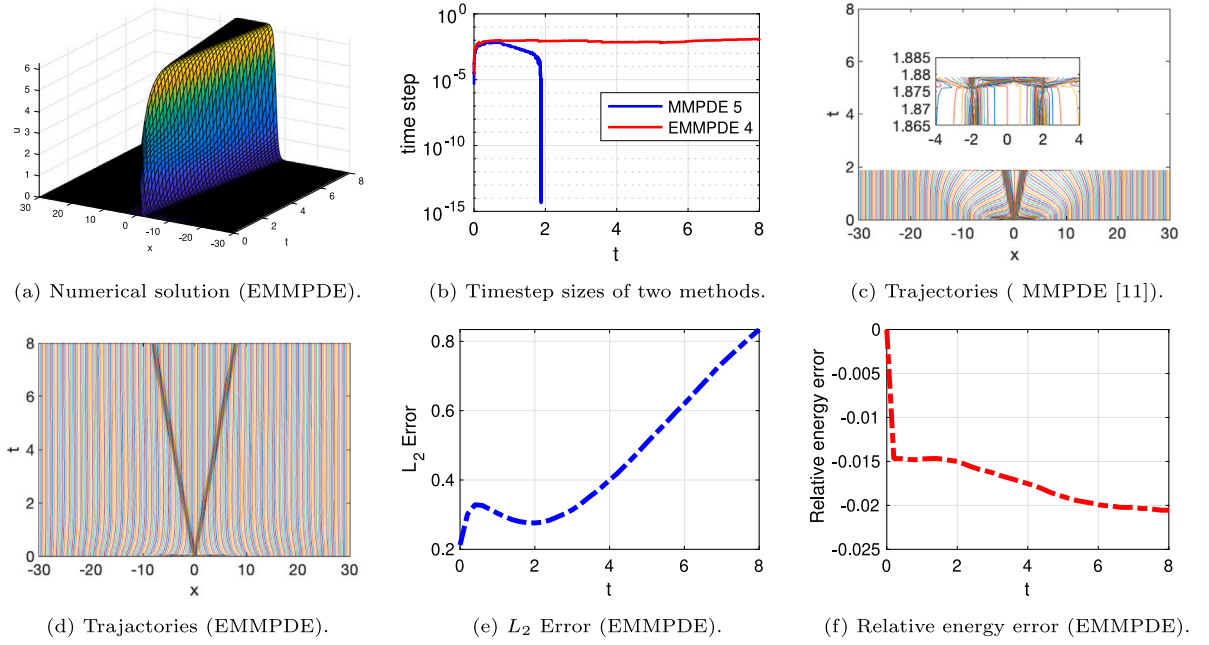


Fig. 1. Numerical experiments applying MMPDE [10] and EMMPDE.

$$u(x, t) = 4 \tan^{-1} \left(\frac{\sinh\left(\frac{\gamma t}{\sqrt{1-\gamma^2}}\right)}{\gamma \cosh\left(\frac{x}{\sqrt{1-\gamma^2}}\right)} \right).$$

The initial conditions are computed using the exact solution at $t = 0$.

This system describes a kink–antikink interaction, where two solitons travel in opposite directions with a speed $c \in (0, 1)$, resulting in two wave fronts propagating away from each other. As c approaches 1, the wave fronts become increasingly steep and move more rapidly, posing significant challenges for simulation.

To showcase the robustness and efficiency of the proposed EMMPDEs, we select arc-length-equidistributed MMPDE 5 [10] as the comparison, which is identified as one of the most efficient MMPDEs: $\frac{\partial x}{\partial t} = \frac{1}{\tau} \frac{\partial}{\partial \xi} \left(M \frac{\partial x}{\partial \xi} \right)$, where $M = \sqrt{1 + u_x^2}$.

Fig. 1 shows the numerical experiments applying two moving mesh strategies for Sine–Gordon equation with $N = 200$. Fig. 1(a) presents the numerical solution obtained with the EMMPDE method. Fig. 1(b) illustrates the time step sizes for both methods, where the MMPDE method failed to continue as T approached 1.8, while the EMMPDE method maintained a stable time step around 10^{-3} throughout the simulation. Fig. 1(c) shows the trajectories applying the MMPDE method, where oscillations and grid entanglement began at around $T = 1.8$. In contrast, Fig. 1(d) shows the trajectories produced by the EMMPDE method, which remained smooth and stable up to $T = 8$. Lastly, Fig. 1(e) and Fig. 1(f) display the L_2 error and relative energy error applying the EMMPDE method, demonstrating strong performance in both error and energy conservation. This example highlights that the proposed EMMPDE method is capable of effectively simulating Hamiltonian PDEs with faster and steeper solitons.

5.2. Two dimensional (2-D) Hamiltonian PDE

We consider the 2-D Sine–Gordon equation

$$u_{tt} = u_{xx} + u_{yy} - \sin(u) \quad (5.2)$$

with the initial condition

$$u(x, y, 0) = 4 \arctan \left[\exp \left(4 - \sqrt{\frac{x^2}{0.436^2} + \frac{(y+7)^2}{0.436^2}} \right) \right] + 4 \arctan \left[\exp \left(4 - \sqrt{\frac{x^2}{0.436^2} + \frac{(y-7)^2}{0.436^2}} \right) \right],$$

$$u_t(x, y, 0) = 4.13 \operatorname{sech} \left(4 - \sqrt{\frac{x^2}{0.436^2} + \frac{(y+7)^2}{0.436^2}} \right) + 4.13 \operatorname{sech} \left(4 - \sqrt{\frac{x^2}{0.436^2} + \frac{(y-7)^2}{0.436^2}} \right),$$

over the domain $\Omega = [-20, 20]^2$. The energy is

$$E(t) = \frac{1}{2} \iint_{\Omega} \left[u_t^2 + u_x^2 + u_y^2 + 2(1 - \cos u) \right] dx dy. \quad (5.3)$$

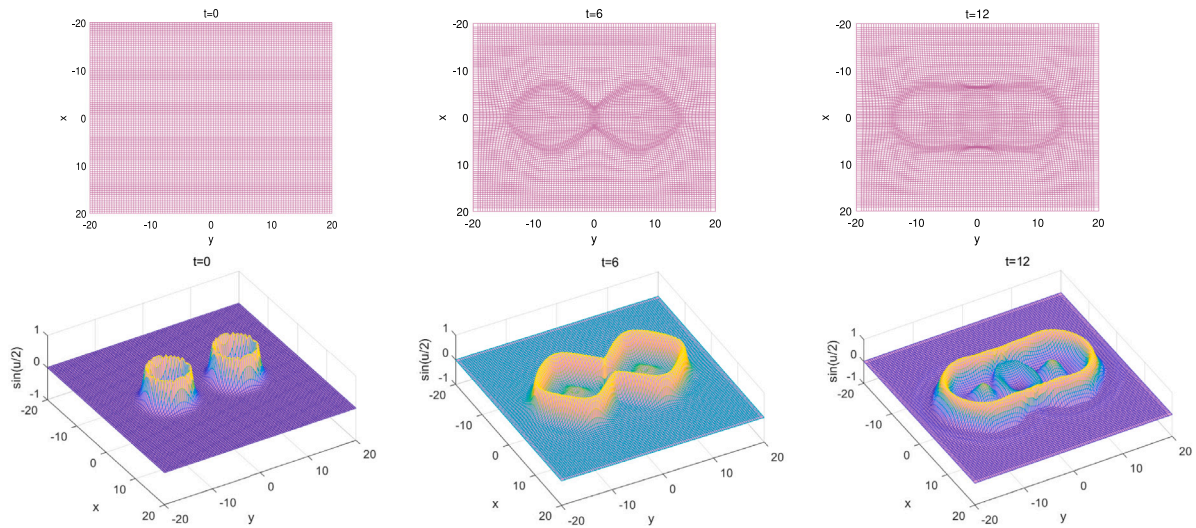


Fig. 2. Mesh movement and numerical simulations applying EIMPDE.

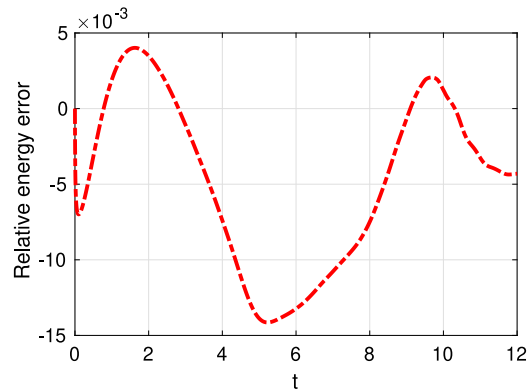


Fig. 3. Relative energy error (EIMPDE).

Eq. (5.2) depicts the interaction of two ring solitons that collide and then expand towards the boundaries. Simulating steep solitons with uniform meshes is challenging.

Using a similar construction technique as in the 1-D EIMPDE, we develop a 2-D moving mesh strategy based on the energy Eq. (5.3). The 2-D EIMPDE is constructed using the variational method described in [19]. We then implement Algorithm 1 from [19] to simulate the 2-D Sine–Gordon equation. The simulation is configured with $N_x = 100$ and $N_y = 100$ spatial grid points, along with a time step of 0.01.

Fig. 2 displays our numerical results in terms of $\sin(u/2)$. At $T = 0$, the initial knots are arranged in a uniform grid. The grid movement and numerical solutions at $T = 6$ and $T = 12$ are also depicted. It is evident that the grid adaptively concentrates in areas of high numerical variation while maintaining smooth and stable transitions. Notably, our simulation uses only a quarter of the grid points compared to [13], yet it covers a longer time interval of $T = 12$, whereas the method in [13] is limited to $T = 11$.

Fig. 3 shows the relative energy error of the system over the time interval from $T = 0$ to $T = 12$. The results demonstrate that our proposed algorithm conserves energy, highlighting its robustness and efficiency.

Remark 5.1. It is noted that these examples have been simulated using various adaptive energy conservation methods, as demonstrated in [2,13]. The success of these methods stems from their integration with energy-conserving algorithms—a significant area of recent research and the focus of our future work. With the proposed EIMPDEs, we are optimistic about developing adaptive energy-conserving algorithms that are more stable and simpler than arc-length based MMPDEs.

Data availability

Data will be made available on request.

References

- [1] K. Feng, M.Z. Qin, *Symplectic Geometric Algorithms for Hamiltonian Systems*, Springer-Verlag, Berlin, 2010.
- [2] S. Eidnes, B. Owren, T. Ringholm, Adaptive energy preserving methods for partial differential equations, *Adv. Comput. Math.* 44 (3) (2018) 815–839.
- [3] B. Cano, Conserved quantities of some Hamiltonian wave equations after full discretization, *Numer. Math.* 103 (2006) 197–223.
- [4] S. Damelin, J. Levesley, D. Ragozin, X. Sun, Energies, group-invariant kernels and numerical integration on compact manifolds, *J. Complexity* 25 (2) (2009) 152–162.
- [5] H.J. Zhu, S.H. Song, Y.F. Tang, Multi-symplectic wavelet collocation method for the nonlinear Schrödinger equation and the Camassa-Holm equation, *Comput. Phys. Comm.* 182 (3) (2011) 616–627.
- [6] L. Brugnano, F. Iavernaro, D. Trigiante, A note on the efficient implementation of Hamiltonian BVMs, *J. Comput. Appl. Math.* 236 (3) (2011) 375–383.
- [7] R. McLachlan, G. Quispel, G. Turner, Numerical integrators that preserve symmetries and reversing symmetries, *SIAM J. Numer. Anal.* 35 (2) (1998) 586–599.
- [8] Z.J. Sun, L. Ling, A kernel-based meshless conservative Galerkin method for solving Hamiltonian wave equations, *SIAM J. Sci. Comput.* 44 (4) (2022) 2789–2807.
- [9] C. de Boor, Good approximation by splines with variable knots, in: *Spline Functions and Approximation Theory: Proceedings of the Symposium Held At the University of Alberta*, Springer, 1973.
- [10] W.Z. Huang, R.D. Russell, *Adaptive Moving Mesh Methods*, vol. 174, Springer Science & Business Media, New York, 2010.
- [11] L.B. Liu, L. Xu, Y.N. Xu, Z.T. Huang, An adaptive grid method for a two-parameter singularly perturbed problem with non-smooth data, *Appl. Math. Lett.* 109200 (2024).
- [12] J.W. Sun, X. Qian, H. Zhang, S.H. Song, Novel energy dissipative method on the adaptive spatial discretization for the Allen-Cahn equation, *Chin. Phys. B* 30 (7) (2021) 070201.
- [13] Z.J. Sun, Q.J. Gao, Energy-preserving schemes for conservative PDEs based on periodic quasi-interpolation methods, *Commun. Nonlinear Sci. Numer. Simul.* 131 (2024) 107831.
- [14] Y.C. Hon, X.Z. Mao, An efficient numerical scheme for Burgers' equation, *Appl. Math. Comput.* 95 (1) (1998) 37–50.
- [15] R. Li, T. Tang, P.W. Zhang, Moving mesh methods in multiple dimensions based on harmonic maps, *J. Comput. Phys.* 170 (2) (2001) 562–588.
- [16] Z.M. Wu, Dynamical knot and shape parameter setting for simulating shock wave by using multi-quadric quasi-interpolation, *Eng. Anal. Bound. Elem.* 29 (4) (2005) 354–358.
- [17] Q.J. Gao, Z.M. Wu, S.G. Zhang, Adaptive moving knots meshless method for simulating time dependent partial differential equations, *Eng. Anal. Bound. Elem.* 96 (2018) 115–122.
- [18] Q.J. Gao, S.G. Zhang, Moving mesh strategies of adaptive methods for solving nonlinear partial differential equations, *Algorithms* 9 (4) (2016) 86.
- [19] Q.J. Gao, S.G. Zhang, Moving mesh method for simulating high-dimensional time dependent PDEs with fast propagating shock waves, *Eng. Anal. Bound. Elem.* 103 (2019) 116–125.